Acceptance Criteria: Android HTML/Markdown Reader

Nov 5, 2026 · @Gonzalo

Scope and decisions

v1 is a read-only viewer for Markdown and HTML files kept in a folder the user picks, with folders, move, rename and delete.

- **Storage:** The app works on real files in the folder the user chooses, so other apps and sync tools see the same files.
- **HTML:** JavaScript is enabled, because AI-generated pages often include interactive explainers such as tabs, diagrams and sliders. Pages run sandboxed, with no network access.
- **File management:** Move, rename, delete and create folder are all in v1.
- **Out of scope:** Editing file content, search, cloud sync inside the app, HTML text scaling settings in v1, and any file type other than Markdown and HTML.

File list

The home screen shows the current folder: subfolders first, then supported files, each group sorted A to Z.

- **FL-1.** Given the folder holds subfolders and files of mixed types, when I open it, then I see subfolders first, then only `.md`, `.markdown`, `.html` and `.htm` files, sorted A to Z ignoring case. Other file types are hidden.
- **FL-2.** Given a list is visible, when I read a row, then it shows the name and a distinct icon for folder, Markdown (`.md`/`.markdown`) or HTML (`.html`/`.htm`). A file row also shows its last-modified date.
- **FL-3.** Given a folder has no subfolders and no supported files, when I open it, then I see an empty state with a short hint.
- **FL-4.** Given files were added, renamed or deleted outside the app, when I return to the app or pull to refresh, then the list matches what is on disk without restarting the app.

Folders and navigation

The user picks one root folder once, and every folder in the app is a real folder inside it.

- **FN-1.** Given the app is opened for the first time, when it starts, then it asks me to choose a root folder before showing any list.
- **FN-2.** Given I chose a root folder earlier, when I reopen the app, then it shows that folder without asking again, and I can change the root folder from settings.
- **FN-3.** Given the root folder is no longer reachable (deleted, or access revoked), when I open the app, then I see a message explaining it and a button to choose a folder again, and the app does not crash.
- **FN-4.** Given I tap a folder, when it opens, then the title or breadcrumb shows the path from the root, and Back goes up one level. Back at the root leaves the app.
- **FN-5.** Given I am in any folder, when I create a folder and give it a valid name, then it appears in the list and exists on disk.
- **FN-6.** Given I enter an empty name, a name that already exists in that folder, or a name with invalid characters, when I confirm, then the app rejects it with a clear message and creates nothing.

Move, rename and delete

I can reorganize files and folders from the list, and nothing is lost or overwritten without my confirmation.

- **MV-1.** Given I long-press a file or folder, when it gets selected, then I can select more items by tapping, and an action bar shows Move, Rename (one item only) and Delete.
- **MV-2.** Given one or more items are selected, when I choose Move and pick a destination folder, then the items appear there and disappear from the origin, on disk as well. If the current working folder is moved, the navigation automatically resets to the nearest parent folder.
- **MV-3.** Given I am choosing a destination, when I look at the folder picker, then I can navigate subfolders, and a folder cannot be moved into itself or into its own subfolders.
- **MV-4.** Given a file with the same name already exists at the destination, when I confirm the move, then I choose between cancel and keep both (the moved item gets a numeric suffix). Nothing is overwritten silently.

- **RN-1.** Given one item is selected, when I choose Rename and enter a valid new name, then the list shows the new name, and a file keeps its extension unless I change it to another supported one.
- **RN-2.** Given I enter an empty name, an existing name in that folder or invalid characters, when I confirm, then the app rejects it with a clear message and the item keeps its old name.
- **DL-1.** Given one or more items are selected, when I choose Delete, then a confirmation dialog states how many items go, and names the folder if a folder with content is included.
- **DL-2.** Given I confirm deletion, when it completes, then the items are removed from the list and from disk. Cancel changes nothing.
- **DL-3.** Given I delete a folder, when I confirm, then the folder and everything inside it is removed, and the dialog has warned me of that beforehand. If the active folder is deleted, navigation returns to the parent folder.
- **MV-5.** Given a move, rename or delete fails (permission, file in use, storage error), when the app reports it, then I see which items failed and why, and the items that succeeded stay done.

Markdown viewer

Tapping a Markdown file shows it fully rendered, never as raw text.

- **MD-1.** Given a Markdown file uses headings, bold, italic, strikethrough, lists, task lists, quotes, horizontal rules, inline code and code blocks, when I open it, then each element is rendered with its own distinct style.
- **MD-2.** Given a file contains tables, when I open it, then the table is rendered with its columns, and a table wider than the screen scrolls horizontally instead of breaking the layout.
- **MD-3.** Given a code block is wider than the screen, when I view it, then it scrolls horizontally and keeps its monospace formatting.
- **MD-4.** Given the file references an image with a relative path, when I open it, then the image loads from the file's own folder. A missing image shows a placeholder, not an error screen.
- **MD-5.** Given the file contains a link to a website, when I tap it, then it opens in the browser.
- **MD-6.** Given the file contains a relative link to another Markdown or HTML file in the folder, when I tap it, then that file opens in the viewer, and pressing Back unwinds the multi-level navigation back-stack to return to the previous file.

- **MD-7.** Given a file is empty or contains invalid Markdown, when I open it, then the viewer shows an empty page or the best-effort rendering, and never crashes.
- **MD-8.** Given a Markdown file is open in the viewer, when it is modified, renamed, or deleted externally, then the viewer displays a non-intrusive warning and returns safely to the file list without crashing.

HTML viewer

Tapping an HTML file shows the page as a browser would, including its JavaScript, inside a sandbox that cannot reach the network or other files.

- **HT-1.** Given an HTML file with inline CSS and inline JavaScript, when I open it, then it renders and its scripts run: tabs switch, sections collapse, sliders and buttons respond.
- **HT-2.** Given the page references local CSS, JavaScript or images by relative path (including relative subdirectories), when I open it, then they load securely from the file's own folder and subfolders using a local WebView scheme loader (`WebViewAssetLoader`/request interceptor).
- **HT-3.** Given a page tries to load a resource from the internet (script, stylesheet, image, font, fetch request), when it renders, then the request is blocked, the rest of the page still shows, and nothing crashes. Works the same with airplane mode on.
- **HT-4.** Given a page script throws an error or runs forever, when I use the page, then the app stays responsive, and Back always leaves the page.
- **HT-5.** Given the page contains a link to a website, when I tap it, then it opens in the browser, not inside the viewer.
- **HT-6.** Given the page contains a relative link to another Markdown or HTML file in the folder, when I tap it, then that file opens in the viewer, and pressing Back unwinds the multi-level navigation back-stack.
- **HT-7.** Given a page script tries to read files outside the file's folder, access the device (camera, location, clipboard) or call app internals, when it runs, then access is denied.
- **HT-8.** Given the page has no viewport tag and is wider than the screen, when I open it, then it fits the width of the screen and can be pinch-zoomed.
- **HT-9.** Given an HTML file is open in the viewer, when it is modified or deleted externally, then the viewer safely notifies the user and exits to the file list.

Reading experience

The app is built for comfortable reading and never changes file content.

- **RD-1.** Given the phone uses light or dark mode, when I open the app, file list, settings, dialogs, or a Markdown file, then all UI surfaces follow the system theme dynamically.
- **RD-2.** Given I open an HTML file in dark mode, when it renders, then the page keeps its own colors, and the app does not force a theme on it.
- **RD-3.** Given I am reading a Markdown file, when I change the font size in settings, then the text resizes immediately and the setting is remembered. HTML font size scaling is out of scope for v1.
- **RD-4.** Given I scrolled partway through a file, when I rotate the phone, switch apps or return from another file, then I come back to the same place.
- **RD-5.** Given I reopen a file I read before, when it loads, then it opens at the top unless I returned within the same session.
- **RD-6.** Given any file is open, when I look for edit controls, then there are none. The viewer never writes to the file.

Non-functional

The app works fully offline, opens files quickly and asks for nothing beyond access to the chosen folder.

- **NF-1.** Given airplane mode is on, when I use any feature, then everything works. The app declares no internet permission.
- **NF-2.** Given a 1 MB Markdown or HTML file on a mid-range phone, when I tap it, then the content is visible in under 1 second.
- **NF-3.** Given a folder with 1,000 items, when I scroll the list, then directory indexing is performed asynchronously on background coroutines/threads using Cursor queries over SAF, remaining smooth and opening in under 1 second without UI jank or ANR.
- **NF-4.** Given the app is installed, when I check its permissions, then the only access it holds is to the folder I picked.
- **NF-5.** Given the app is running, when I inspect it, then it has no ads, analytics or tracking.
- **NF-6.** Given a device on Android 8.0 (API 26) or newer, when I install and use the app, then every feature in this document works.

- **NF-7.** Given a file larger than 10 MB, when I tap it, then the app opens it, shows a loading state while parsing, enforces memory-safe streaming/AST limits to prevent `OutOfMemoryError`, and does not freeze or crash.